Regulatory Separation and Pharmaceutical Innovation: Evidence from China's MAH Reform

Abstract

This paper develops a partial-equilibrium mechanism model of how China's Marketing Authorization Holder (MAH) system changes the value of advancing viable drug projects and their commercialization organization. Developers choose one common original-drug innovation investment / project-advancement intensity before project characteristics and organizational routes are realized. MAH makes retained entrusted manufacturing legally available while the developer retains authorization-holder responsibility. Developers also differ in predetermined financing capacity for commercialization commitments. Internal production and retained entrusted production require distinct up-front commitments, but those commitments are liquidity requirements rather than additional route costs. MAH does not expand financing supply. It reduces the amount of vertical-integration capital that must be financed to retain commercialization rights, and is therefore especially valuable when a developer cannot finance full internalization but can finance retained entrusted production. Route choice remains deterministic, and the price of qualified manufacturing services is endogenous. Reform effects are confined to project–developer pairs for which the new route is financeable and improves on the best old option; service scarcity can attenuate these gains. The baseline does not treat MAH as a demand, credit-supply, scientific-productivity, downstream-success, or patenting shock.

Keywords: MAH; pharmaceutical innovation; commercialization frictions; complementary assets; entrusted production; financing capacity.
JEL codes: D22; L65; O31; O38; I18.

1 Introduction

China's MAH system separates the right to hold marketing authorization from a strict requirement of in-house production. This institutional change is potentially important for drug development because many research-originating firms can generate valuable projects but lack complete manufacturing capacity. The central mechanism studied here is therefore not demand expansion or scientific productivity. It is an expected-commercialization-value channel: when a firm anticipates that an advancing project can later use qualified external manufacturing while the firm retains authorization, the expected value of advancing that project may increase. The reform can be particularly important when weak internal manufacturing capability makes vertical integration capital intensive. MAH does not relax the firm's financing constraint; it changes the organizational technology required to retain commercialization rights.

The mechanism builds on two established ideas. First, pharmaceutical innovation responds to expected downstream rewards (Schmookler, 1966; Acemoglu and Linn, 2004; Finkelstein, 2004; Dubois et al., 2015; Budish et al., 2015). Second, an innovator's ability to appropriate returns depends on complementary assets, firm boundaries, and access to specialized inputs (Teece, 1986; Arora et al., 2001; Gans and Stern, 2003; Grossman and Helpman, 2002; Bøler et al., 2015; Ma, 2026). Financing frictions can make large up-front innovative investments difficult to fund (Hall and Lerner, 2010). MAH combines these ideas narrowly: it changes how an innovator can access manufacturing capability while retaining the authorization asset, without changing the supply of finance.

Recent evidence on China's pharmaceutical reforms helps separate this route mechanism from two nearby channels. Jia et al. (2023) study the 2015 drug-review reform and emphasize approval delay, review capacity, and the composition of firms and products entering the pipeline. Barwick et al. (2026) study the National Reimbursement Drug List and emphasize demand and market-size incentives. Those channels can coexist with MAH, but they are not the MAH mechanism in this paper. The baseline holds scientific productivity and the market-return environment fixed and asks whether a regulated holder–producer separation route changes expected commercialization value.

Closely related evidence also shows why project advancement and upstream patenting must remain distinct. Gu (2024) finds that allowing production outsourcing raises clinical-trial registrations among innovators without production facilities, while patent applications fall among firms with fewer financial resources. The paper also reports heterogeneous responses across development stages and drug classes. Gu studies financing-induced reallocation between research and development. The present baseline instead allows financing capacity to determine whether a developer can fund vertical integration or the smaller organizational commitment required for retained entrusted production. This distinction is a maintained model boundary, not a claim of empirical novelty: MAH changes route feasibility and value here, but does not directly constrain the advancement control or generate patents.

For a development-economics audience, the relevant bottleneck is a regulated complementary asset. MAH can relax a manufacturing bottleneck for research-oriented firms while preserving holder responsibility. Scarcity in qualified production support can still attenuate the effect.

The model is deliberately narrow. It is not a full dynamic structural model of the pharmaceutical industry. It does not solve product-market clearing, an input market for project advancement, free entry, exit, or an invariant firm distribution. The baseline closes only the market directly tied to the MAH mechanism: qualified manufacturing services. Reform-induced demand for retained entrusted production can raise the equilibrium service price p_m^*, so the model does not mechanically imply a positive effect for every project or developer.

The contribution is to organize the mechanism in a way that is internally consistent with feasible data. The theory separates project advancement from planning-stage project arrival, deterministic organizational assignment, observed holder–producer separation, and realized products. The empirical role of the model is correspondingly modest: it specifies empirically interpretable interfaces for project-level organization, retained outcomes, and service scarcity without claiming that approval-side data separately identify every primitive friction.

The main text gives the shortest complete commercialization baseline. The technical appendix records primitives, financing and payoff accounting, pricing derivations, deterministic sorting, project advancement, qualified-capacity equilibrium, comparative statics, observed outcomes, and explicitly separated inactive extensions.

2 Institutional Background

The relevant institutional point is timing. Under China's drug-registration framework, a firm need not already be the final commercial holder when it begins to advance a viable project. The economic choice is forward-looking: a firm invests today while anticipating the manufacturing and commercialization environment it will face if a project advances. Production support and manufacturing readiness must be planned before final approval is observed. After obtaining a Drug Approval License, the applicant becomes the MAH (National Medical Products Administration, 2022a). The model therefore treats MAH as changing the anticipated availability and value of a future organizational route, not current scientific productivity or downstream success.

The MAH reform creates a regulated holder-producer separation route. The 2016 pilot allowed eligible applicants in selected regions to become holders under the pilot arrangement (General Office of the State Council of the People's Republic of China, 2016; China Food and Drug Administration, 2016). The later legal framework consolidated the principle that an MAH may manufacture by itself or entrust production to a qualified manufacturer (National Medical Products Administration, 2019). Subsequent NMPA rules require holder-side quality responsibility, assessment of the entrusted manufacturer, quality and entrustment agreements, and ongoing supervision (National Medical Products Administration, 2022c,b, 2023).

These rules imply two modeling restrictions. First, entrusted production is not anonymous outsourcing or transfer. It is a regulated retained route in which authorization remains with the developer while production is performed by a qualified external manufacturer. Second, MAH does not eliminate holder responsibility. The entrusted payoff therefore retains a holder-side burden μ_E, while the institutional wedge $\tau_E(M)$ governs effective route availability. The downstream realization probability $s(q)$ is route independent and invariant to MAH in the baseline.

3 Model

3.1 Environment and timing

There is a continuum of developers. Developer i has predetermined project-advancement capability $a_i > 0$, internal manufacturing capability $k_i > 0$, and commercialization-stage financing capacity $\ell_i > 0$:

$$\theta_i = (a_i, k_i, \ell_i) \sim H(a, k, \ell). \tag{1}$$

The joint distribution permits arbitrary correlation. Financing capacity is predetermined and measures the largest up-front organizational commitment the developer can fund; it is distinct from both research capability and manufacturing know-how. A viable project that reaches commercialization-relevant planning draws value $q > 0$ and manufacturing requirement $m > 0$ from the exogenous distribution $F(q, m)$. For empirical decompositions only,

$$F(q, m) = \sum_{g \in \{O, \mathrm{Inc}\}} \rho_g F_g(q, m), \qquad \rho_g \geq 0, \quad \sum_g \rho_g = 1.$$

The class label g creates no class-specific control.

The institutional regime is $M \in \{0, 1\}$. The pre-MAH regime sets the barrier to retained entrusted manufacturing at infinity; the MAH regime makes it finite:

$$\tau_E(0) = +\infty, \qquad \tau_E(1) = \bar{\tau}_E < +\infty, \qquad \bar{\tau}_E \geq 0. \tag{2}$$

The available routes are internal manufacturing I, retained entrusted manufacturing E, non-retained transfer T, and abandonment or delay A. Under E, the developer remains the authorization holder and a qualified external producer manufactures. Thus E is not transfer.

Before project characteristics and routes are realized, developer i chooses one common $x_i \geq 0$, interpreted as original-drug innovation investment / project-advancement intensity. It is broader than pure clinical development, but excludes basic research, compound discovery, patent-generating effort, and patent applications. The expected measure of viable projects reaching route planning is

$$\lambda_i^{\text{plan}} = a_i x_i. \tag{3}$$

Timing is as follows. First, M is observed and developers anticipate the market-consistent qualified-service price. Second, after observing (a_i, k_i, ℓ_i, M), they choose x_i. Third, planning-stage projects draw (q, m); the developer checks route financing requirements against ℓ_i and chooses among the financeable routes. Fourth, downstream realization occurs with exogenous probability $s(q)$. Finally, aggregate capacity demand and supply must validate the anticipated service price. The last step is an equilibrium consistency condition, not a late unanticipated shock. There is no financing decision or finance market. Upstream research and patent generation may help determine predetermined a_i, but they are outside the baseline decision problem.

3.2 Commercialization technologies

Conditional on successful commercialization, residual demand is $y(p; q) = Aqp^{-\varepsilon}$, where $A > 0$ and $\varepsilon > 1$. Product-price optimization gives

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1} c$$

and the gross operating present value

$$R(q, c) = \frac{Aq}{1 - \beta\varphi} \frac{(\varepsilon - 1)^{\varepsilon - 1}}{\varepsilon^{\varepsilon}} c^{1-\varepsilon}, \qquad \beta \in (0, 1), \quad \varphi \in [0, 1). \tag{4}$$

The appendix derives the FOC, SOC, and accounting. In particular, $R_q > 0$ and $R_c < 0$.

Internal production uses marginal cost $c_I(m, k_i)$, with $c_{I,m} > 0$ and $c_{I,k} < 0$, and setup cost $F_I(m, k_i)$, with $F_{I,m} > 0$ and $F_{I,k} < 0$. Both are finite for $k_i > 0$. Its economic route value is

$$W_i^I(q, m) = s(q) R(q, c_I(m, k_i)) - F_I(m, k_i). \tag{5}$$

Internalization requires an up-front financing commitment $J_I(m, k_i) > 0$, where

$$J_{I,m} > 0, \qquad J_{I,k} < 0.$$

The commitment is a liquidity requirement, not an additional real cost: F_I is subtracted once in W_i^I, whereas J_I only determines whether that route can be funded.

Entrusted production uses external marginal cost $c_E(m)$, requires $b(m) > 0$ units of qualified capacity with $b'(m) > 0$, incurs setup cost $F_E(m)$, and leaves the holder with burden $\mu_E \geq 0$. At the qualified-capacity price p_m,

$$\begin{aligned} W_i^E(q, m; M, p_m) = s(q)R(q, c_E(m)) - F_E(m) - p_m b(m) \\ - \mu_E - \tau_E(M). \end{aligned} \tag{6}$$

Retained entrusted production requires its own positive up-front commitment $J_E(m)$, with $J_E'(m) > 0$. This is also a liquidity requirement and is not subtracted from W_i^E. Each real monetary cost therefore enters once. The transfer and abandonment values are

$$W^T(q, m) = T(q, m), \qquad W^A = 0. \tag{7}$$

Financing-adjusted route values are

$$\widetilde{W}_i^I(q, m; \ell_i) = \begin{cases} W_i^I(q, m), & J_I(m, k_i) \leq \ell_i, \\ -\infty, & J_I(m, k_i) > \ell_i, \end{cases} \tag{8}$$

and

$$\widetilde{W}_i^E(q, m; M, p_m, \ell_i) = \begin{cases} W_i^E(q, m; M, p_m), & J_E(m) \leq \ell_i, \\ -\infty, & J_E(m) > \ell_i. \end{cases} \tag{9}$$

MAH changes none of ℓ_i, J_I, J_E. Pre-MAH unavailability of E continues to arise from $\tau_E(0) = +\infty$, not from financing.

3.3 Organization and project advancement

At a fixed p_m, the project chooses deterministically:

$$\begin{aligned} \widetilde{W}_i(q, m; M, p_m, \ell_i) = \max\{\widetilde{W}_i^I, \widetilde{W}_i^E, W^T, W^A\}, \\ r_i^*(q, m; M, p_m, \ell_i) \in \arg\max_{r \in \{I,E,T,A\}} \widetilde{W}_i^r. \end{aligned} \tag{10}$$

Continuous heterogeneity makes ties a zero-measure event; no logit or inclusive value is used.

The internal-minus-entrusted gap is

$$\begin{aligned} \Delta_{IE}(k_i) = s(q)[R(q, c_I(m, k_i)) - R(q, c_E(m))] - F_I(m, k_i) \\ + F_E(m) + p_m b(m) + \mu_E + \tau_E(M). \end{aligned} \tag{11}$$

Holding (q, m, M, p_m) fixed on the internally feasible domain,

$$\Delta_{IE,k} = s(q) R_c(q, c_I)\, c_{I,k} - F_{I,k} > 0.$$

Proposition 1 (Organizational feasibility and sorting)**.** *Route I is financially feasible if and only if $J_I(m,k_i) \le \ell_i$; route E is financially feasible if and only if $J_E(m) \le \ell_i$, in addition to its institutional availability. Since $J_{I,k} < 0$, stronger internal capability weakly expands the set of projects for which internalization can be financed. Conditional on both retained routes being financeable, on both beating T and A, and on the endpoint crossing conditions, there is a unique value cutoff $k^*(q,m;p_m,M)$ satisfying $\Delta_{IE}(k^*) = 0$. Developers below it select E, and those above it select I.*

The financeability screen precedes value ranking; comparative-advantage sorting remains. In the empirically relevant low-capability region, impose the local condition

$$J_E(m) < J_I(m,k_i).$$

It says that retained outsourcing requires less up-front financing than building internal capacity and is not a global restriction.

Define the best pre-MAH value and post-MAH value by

$$\begin{aligned} W_i^0(q,m;\ell_i) &= \max\{\widetilde{W}_i^I(q,m;\ell_i), W^T(q,m), 0\}, \\ W_i^1(q,m;p_m,\ell_i) &= \max\{W_i^0(q,m;\ell_i), \widetilde{W}_i^E(q,m;1,p_m,\ell_i)\}. \end{aligned} \tag{12}$$

The financing-adjusted MAH-relevant set is

$$\mathcal{C}_i(p_m,\ell_i) = \Big\{(q,m) : J_E(m) \le \ell_i, \quad W_i^E(q,m;1,p_m) > W_i^0(q,m;\ell_i)\Big\}. \tag{13}$$

Proposition 2 (MAH-relevant projects and the financing corridor)**.** *At a fixed p_m,*

$$W_i^1 - W_i^0 = [\widetilde{W}_i^E - W_i^0]_+, \tag{14}$$

so a strict project-value gain occurs exactly on $\mathcal{C}_i(p_m,\ell_i)$. Suppose locally that $J_E(m) < J_I(m,k_i)$. If $\ell_i < J_E(m)$, neither retained route is financeable and MAH has no effect. If $J_E(m) \le \ell_i < J_I(m,k_i)$, only retained entrusted production is financeable, and the gain is strict exactly when $W_i^E > \max\{W^T, 0\}$. If $\ell_i \ge J_I(m,k_i)$, both retained routes are financeable and the gain is $[W_i^E - \max\{W_i^I, W^T, 0\}]_+$. The reform effect is therefore not globally monotone in financing capacity.

Expected optimized value per planning-stage project is

$$\Omega_i(M,p_m) = \int \widetilde{W}_i(q,m;M,p_m,\ell_i)\, dF(q,m). \tag{15}$$

Developer i solves

$$\max_{x_i \ge 0} \left\{\beta a_i x_i \Omega_i(M,p_m) - \frac{\kappa}{1+\nu} x_i^{1+\nu}\right\}, \qquad \kappa > 0, \quad \nu > 0. \tag{16}$$

Strict concavity gives the unique solution

$$x_i^*(M,p_m) = \left[\frac{\beta a_i}{\kappa}\Omega_i(M,p_m)\right]^{1/\nu}, \tag{17}$$

including $x_i^* = 0$ when $\Omega_i = 0$. The binary reform affects x_i^* only through Ω_i. At a fixed p_m, its effect is strict only if E improves on the pre-MAH maximum on a positive-probability project set.

Proposition 3 (Project advancement and heterogeneity)**.** *At a fixed p_m, the reform strictly raises x_i^* if and only if*

$$\int_{\mathcal{C}_i(p_m,\ell_i)} [W_i^E(q,m;1,p_m) - W_i^0(q,m;\ell_i)]\, dF(q,m) > 0. \tag{18}$$

Research capability a_i scales any positive response. Since $J_{I,k} < 0$, lower k_i makes the financing corridor more likely at a given ℓ_i, but it does not by itself guarantee a treatment effect. Within either regime, a larger ℓ_i weakly expands the feasible route set and hence weakly raises x_i^*. The reform response $\Delta x_i(\ell_i)$, however, has no global monotonic sign because greater financing capacity also restores internalization as a pre-MAH option.

3.4 Qualified manufacturing-service equilibrium

Qualified supplier j, with efficiency z_j, chooses capacity $s_j \geq 0$:

$$\max_{s_j\geq 0}\{p_m s_j - \Psi(s_j;z_j)\}. \tag{19}$$

With $\Psi_{ss} > 0$, positive-price capacity is uniquely determined by $p_m = \Psi_s(s_j^*;z_j)$, and aggregate supply is

$$S_m(p_m) = \int s_j^*(p_m,z)\, dH_C(z).$$

Expected entrusted capacity per planning-stage project is

$$\chi_i^E(p_m;M) = \int b(m)\mathbf{1}\{r_i^*(q,m;M,p_m,\ell_i) = E\}\, dF(q,m). \tag{20}$$

Study-related and total demand are

$$\begin{aligned} D_m^{\mathrm{MAH}}(p_m;M) &= \int a_i x_i^*(M,p_m)\chi_i^E(p_m;M)\, dH(a,k,\ell), \\ D_m(p_m;M) &= D_m^B(p_m) + D_m^{\mathrm{MAH}}(p_m;M). \end{aligned} \tag{21}$$

The demand expression includes both feedbacks: price changes expected value and hence advancement, and it changes deterministic route selection. Continuous heterogeneity makes aggregate demand continuous even though individual indicators can jump.

The qualified-capacity market clears at the unique price

$$D_m(p_m^*; M) = S_m(p_m^*). \tag{22}$$

Supply is strictly increasing and demand weakly decreasing. Positive background demand gives excess demand at zero, while entrusted and background demand vanish and supply diverges at high prices. These conditions give existence and uniqueness. If post-MAH study demand is positive at the old price, $p_m^*(1) > p_m^*(0)$: CMO scarcity attenuates the reform rather than amplifying it through a direct price reduction.

For the equilibrium comparison, let $p_m^h \equiv p_m^*(h)$, and define

$$\Delta\Omega_i^{\text{dir}} \equiv \Omega_i(1, p_m^0) - \Omega_i(0, p_m^0), \qquad \Delta\Omega_i^{\text{eq}} \equiv \Omega_i(1, p_m^1) - \Omega_i(0, p_m^0), \tag{23}$$

$$\Delta x_i^{\text{dir}} \equiv x_i^*(1, p_m^0) - x_i^*(0, p_m^0), \qquad \Delta x_i^{\text{eq}} \equiv x_i^*(1, p_m^1) - x_i^*(0, p_m^0). \tag{24}$$

Proposition 4 (CMO equilibrium and scarcity attenuation)**.** *Under atomless financing and route heterogeneity, the existing integrability conditions, weakly decreasing demand, and strictly increasing supply, the scalar CMO market has a unique clearing price. Moreover,*

$$p_m^*(1) \geq p_m^*(0), \qquad 0 \leq \Delta\Omega_i^{\text{eq}} \leq \Delta\Omega_i^{\text{dir}}, \qquad 0 \leq \Delta x_i^{\text{eq}} \leq \Delta x_i^{\text{dir}}. \tag{25}$$

The price inequality is strict when post-MAH study demand is positive at the old price. Financing capacity shifts neither CMO supply nor the background demand schedule.

The baseline partial equilibrium contains exactly

$$\left\{p_m^*,\ x_i^*,\ r_i^*(q, m),\ s_j^*\right\}_{i,j}. \tag{26}$$

No entry, labor, capital, product-market aggregate, or welfare closure is added.

3.5 Observed outcomes and nesting

Planning, organization, and observed outcomes. At equilibrium,

$$\Lambda_i^{\text{plan}} = a_i x_i^*,$$

while expected realized retained outcomes are

$$Y_i^{\text{ret}} = a_i x_i^* \int s(q) \left[\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}\right] dF(q, m). \tag{27}$$

Thus observed outcomes can change through advancement or route composition. Holder–producer separation is observed only after route assignment and cannot cause the earlier x_i choice.

Novelty composition. For $g \in \{O, \mathrm{Inc}\}$, outcomes use the same x_i^* and the mixture $\rho_g F_g$. The baseline permits either class to have a zero or larger response and imposes no original-versus-incremental ranking. Patent applications are not a baseline endogenous outcome.

No-binding-finance limit. If

$$\ell_i \geq \max\{J_I(m, k_i), J_E(m)\}$$

for every relevant developer–project pair, financing never binds, $\widetilde{W}_i^I = W_i^I$ and $\widetilde{W}_i^E = W_i^E$. The model then collapses to the previous no-financing-friction commercialization baseline.

4 Empirical Predictions and Mapping Boundaries

The deterministic model yields three conditional predictions. First, among projects for which internal and entrusted production dominate transfer and abandonment, developers with weaker internal manufacturing capability are more likely to select retained entrusted production. This tendency is strongest when pre-policy financing capacity lies between the entrusted and internal up-front requirements. Second, MAH raises project advancement only when the entrusted route is financeable and exceeds the best pre-reform option; the response can be zero when financing is extremely scarce, when internalization is already attractive and financeable, or when an outside option dominates. Third, reform-induced demand for qualified manufacturing services can raise p_m^*, attenuating fixed-price gains in expected project value and advancement. These predictions concern commercialization organization and project advancement, not patent generation or scientific productivity.

The empirical distinction is between ex ante advancement and downstream observations. Product-level holder–manufacturer separation can proxy route E only when holder identity, manufacturer identity, product identity, and effective date are aligned. Approval or launch records are realized-product outcomes, not direct observations of x_i. Clinical milestones may proxy project advancement only when their status and timing are measured consistently. Patent applications are a separate patenting margin and are not a baseline endogenous outcome. They may be studied as a separate empirical outcome, but the present mechanism neither maps them to x_i nor assigns their reform response a sign.

The timing restriction can be summarized as

$$M \longrightarrow \text{anticipated availability and value of } E \longrightarrow \Omega_i \longrightarrow x_i^*, \tag{28}$$

Table 1: Model Objects and Measurable Empirical Interfaces

Model object	Candidate interface	Claim boundary
Predetermined capability a_i	Pre-policy development history or other validated capability measure	Does not separately identify a_i, and no policy shift in a_i is allowed.
Advancement x_i and $a_i x_i$	Consistently measured development-stage activity and planning-stage project events	x_i is not pure clinical effort, patent applications, or an approval count.
Predetermined financing capacity ℓ_i	Validated pre-policy registered or paid-in capital, liquid assets, leverage or debt capacity, listed status, VC/PE or parent backing, or external-financing history	A proxy must predate reform exposure; it does not identify a finance market or imply that MAH changes financing supply.
Route $r_i^* = E$	Product-level holder differs from manufacturer, with time-consistent identifiers	The observed split is downstream evidence on retained organizational assignment, not the policy treatment itself.
Retained outcome Y_i^{ret}	Approval or launch with developer/holder identity	Combines advancement, project mix, deterministic route assignment, and $s(q)$; it does not identify these primitives separately.
CMO scarcity p_m^*	Service price, or consistently measured capacity, utilization, density, waiting-time, or participation proxy	A nonprice proxy does not identify the price or the supply and demand schedules separately.
Novelty class g	Versioned regulatory classification O or Inc	The classifier creates no x_{ig} and imposes no response ranking.
Internal financing requirement $J_I(m, k_i)$	Validated up-front capital needed to build, acquire, validate, or upgrade compliant internal production	It is a liquidity threshold, not an extra route cost; no data source is asserted available before audit.
Entrusted financing requirement $J_E(m)$	Validated up-front transfer, validation, contracting, and holder-side preparation commitment	It is positive in the baseline and distinct from $F_E, p_m b(m), \mu_E$; no data source is asserted available before audit.

followed by

$$
\begin{aligned}
x_i^* &\longrightarrow \text{planning-stage projects} \longrightarrow r_i^* \\
&\longrightarrow \text{observed holder–producer separation} \longrightarrow \text{realized products.}
\end{aligned}
\tag{29}
$$

Observed separation cannot be placed before or treated as a cause of the ex ante x_i choice. The original and incremental classes use the same x_i and the mixture $\rho_g F_g$; either class may have a zero or larger response.

These entries are interfaces for future validated data. No data set is asserted here to be available, complete, or identifying. Before use, a source must pass provenance, coverage, key-uniqueness, unit, timing, and construct-validity checks. Entry, route-specific realization, and smooth route choice remain undeveloped extensions. The primary financing prediction is the triple interaction between MAH exposure, weak internal manufacturing capability, and limited or intermediate pre-policy financing capacity, with retained holder–manufacturer separation as the first outcome.

5 Conclusion

MAH changes expected project value by making retained entrusted manufacturing available while preserving authorization-holder responsibility. The route is most relevant when a developer can advance valuable projects but lacks project-compatible internal manufacturing capability. Predetermined financing capacity adds a feasibility margin: very low-capacity developers may be unable to fund either retained route, intermediate-capacity developers may finance entrusted production but not vertical integration, and high-capacity developers recover internalization as an old option. MAH therefore changes the organizational technology of commercialization; it does not relax the financing constraint itself. Its equilibrium value also depends on qualified-service scarcity: reform-induced entrusted demand can raise p_m^* and attenuate the fixed-price gain. Project advancement rises only for the financing-adjusted MAH-relevant set and need not rise for every developer or novelty class. Observed holder–producer separation and realized products occur after advancement and organizational assignment. Patent applications and upstream scientific research remain outside the baseline, and no patent response is imposed. The model makes no welfare claim.

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